

The empirical recurrence for OEIS A236793: recovering a conjecture that is not written down

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Abstract

OEIS A236793 records “Empirical recurrence of order 83”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 190$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236793 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum plus the upper median plus the minimum of every 2×2 subblock equal.”. It has offset 1 and begins

825, 6217, 47831, 420133, 3722333, 34857701, 327044595, 3142571647, ...

The entry states:

Empirical recurrence of order 83 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:33:30, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 190$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 190$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 380$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	17	c_{22}	−116340952161916838	c_{43}	154113240952271945038052	c_{64}	−62487498
c_2	149	c_{23}	−3272779809783538	c_{44}	36243384479322799503009	c_{65}	−2704236
c_3	−3822	c_{24}	954302011875230349	c_{45}	−252558759358735794910697	c_{66}	16066269
c_4	−5099	c_{25}	−320548644721837094	c_{46}	−12090990432014794852664	c_{67}	−1003221
c_5	385130	c_{26}	−6478914470291856511	c_{47}	348668526245119369378700	c_{68}	−3123273
c_6	−470186	c_{27}	4170768349052004296	c_{48}	−38616999797218798816301	c_{69}	48944503
c_7	−23210949	c_{28}	36612702131063326183	c_{49}	−404308939215042828075753	c_{70}	43923239
c_8	60570417	c_{29}	−33574882254880366927	c_{50}	100905470905950208406915	c_{71}	−1108217
c_9	940504129	c_{30}	−172831891596991986931	c_{51}	391915280100857250940160	c_{72}	−410464
c_{10}	−3379677366	c_{31}	202795687958598677053	c_{52}	−147383978379028506438470	c_{73}	1549756
c_{11}	−27246699520	c_{32}	682536774766893321784	c_{53}	−315346851989417002084179	c_{74}	2014670
c_{12}	121431917218	c_{33}	−975834084459192238409	c_{54}	156919130648950916587258	c_{75}	−135201
c_{13}	584194818792	c_{34}	−2252938899441608549659	c_{55}	208477619911831767937308	c_{76}	150040
c_{14}	−3139393787896	c_{35}	3843391626158026393232	c_{56}	−129927927510832504586596	c_{77}	667504
c_{15}	−9425097731306	c_{36}	6190401975549878168591	c_{57}	−111524272768219126399088	c_{78}	−7479
c_{16}	61598162124697	c_{37}	−12570611244671849135022	c_{58}	85511507912413637949661	c_{79}	−1337
c_{17}	114219500453427	c_{38}	−14030566709588819286664	c_{59}	47101637328217152670628	c_{80}	30750
c_{18}	−946862834116793	c_{39}	34427622778574228358699	c_{60}	−44993104485233133278692	c_{81}	−785
c_{19}	−1004826003316698	c_{40}	25747525903434993537460	c_{61}	−15004974889836196172242	c_{82}	−215
c_{20}	11646036448147900	c_{41}	−79324604036960951397983	c_{62}	18870811452105240611561	c_{83}	141
c_{21}	5571787380506985	c_{42}	−36717985022501057720706	c_{63}	3225918607639655186313		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83 + 4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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